Lecture 10: The Basic Rules of Counting

Section 6.1

Lecture Summary

- The Product Rule
- The Sum Rule
- The Subtraction Rule

Basic Counting Principles: The Product Rule

The Product Rule : A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

Example : How many bit strings of length seven are there?

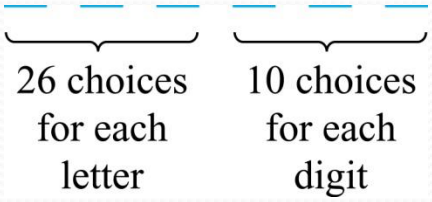
Solution : Since each of the seven bits is either a 0 or a 1, the answer is $2^7 = 128$.

The Product Rule

Example : How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits?

Solution : By the product rule,

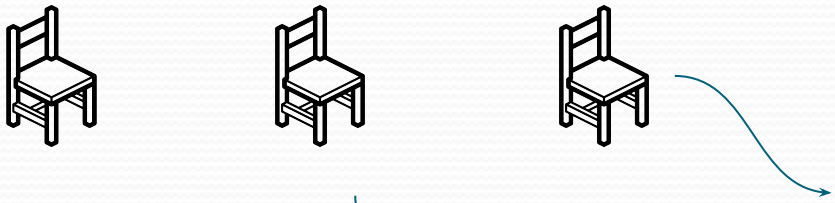
there are $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17{,}576{,}000$ different possible license plates.



Seating Arrangement

Example : We have 5 students, how many different seating arrangements are possible if we have 3 seats available?

Possible solution:



For the first chair, we have 5 students available to choose from

For the second chair, we have 4 students available now to choose from

For the third chair, we have remaining 3 students to choose from

and all three events have to occur all at once, so we can use product rule.
And possible arrangement would be $(5 \times 4 \times 3) = 60$

We can later try to solve the same problem with permutation or combination concept.

Telephone Numbering Plan

Example : The *North American numbering plan* (*NANP*) specifies that a telephone number consists of 10 digits, consisting of a three-digit area code, a three-digit office code, and a four-digit station code. There are some restrictions on the digits.

- Let X denote a digit from 0 through 9.
- Let N denote a digit from 2 through 9.
- Let Y denote a digit that is 0 or 1.

- In the old plan (in use in the 1960s) the format was *NYX-NNX-XXXX*.
- In the new plan, the format is *NXX-NXX-XXXX*.

How many different telephone numbers are possible under the old plan and the new plan?

Solution : Use the Product Rule.

- There are $8 \cdot 2 \cdot 10 = 160$ area codes with the format *NYX.*
- There are $8 \cdot 10 \cdot 10 = 800$ area codes with the format *NXX.*
- There are $8 \cdot 8 \cdot 10 = 640$ office codes with the format *NNX.*
- There are $10 \cdot 10 \cdot 10 \cdot 10 = 10{,}000$ station codes with the format *XXXX.*
  - Number of old plan telephone numbers: $160 \cdot 640 \cdot 10{,}000 = 1{,}024{,}000{,}000$.
  - Number of new plan telephone numbers: $800 \cdot 800 \cdot 10{,}000 = 6{,}400{,}000{,}000$.

Counting Subsets of a Finite Set

Counting Subsets of a Finite Set : Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$.

Solution :
When the elements of S are listed in an arbitrary order, there is a one-to-one correspondence between **subsets of S** and **bit strings of length $|S|$**, when the i th element is in the subset, the bit string has a 1 in the i th position and a 0 otherwise.

By the product rule, there are $2^{|S|}$ such bit strings, and therefore $2^{|S|}$ subsets.

Product Rule in Terms of Sets

- If $A_1, A_2, \ldots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements of each set.
- The task of choosing an element in the Cartesian product $A_1 \times A_2 \times \cdots \times A_m$ is done by choosing an element in A_1, an element in A_2 , …, and an element in A_m.
- By the product rule, it follows that:
  - $|A_1 \times A_2 \times \cdots \times A_m| = |A_1| \cdot |A_2| \cdot \cdots \cdot |A_m|$.

Basic Counting Principles: The Sum Rule

The Sum Rule : If a task can be done **either** in one of n_1 ways **or** in one of n_2, where none of the set of n_1 ways is the **same** as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Example : The mathematics department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student.

Solution : By the sum rule it follows that there are

$37 + 83 = 120$ possible ways to pick a representative.

The Sum Rule in terms of sets

- The sum rule can be phrased in terms of sets.

  $|A \cup B| = |A| + |B|$ as long as A and B are disjoint sets.

- Or more generally,

  $$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$$

  when $A_i \cap A_j = \emptyset$ for all i, j.

- The case where the sets have elements in common will be discussed when we consider the subtraction rule (remember inclusion-exclusion principle?)

Combining the Sum and Product Rule

Example : Suppose variable names in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible names.

Solution : Two ways to make a label.

Only one letter = 26,

1 Letter followed by a digit = 26*10 = 260 *[product rule]*

Total labels = 26 + 260 = 286 *[sum rule]*

Example: Counting Passwords

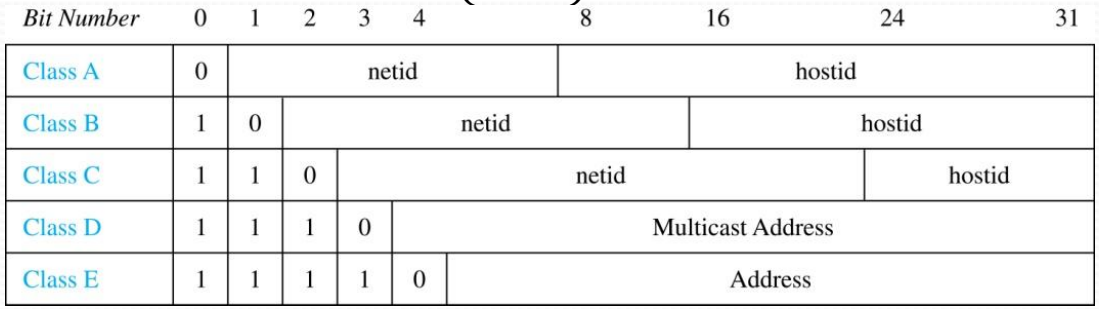
Example : Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution : Let P be the total number of passwords, and let P_6, P_7, and P_8 be the passwords of length 6, 7, and 8.

- By the sum rule $P = P_6 + P_7 + P_8$.
- To find each of P_6, P_7, and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters. We find that:
  - $P_6 = 36^6 - 26^6 = 2{,}176{,}782{,}336 - 308{,}915{,}776 = 1{,}867{,}866{,}560.$
  - $P_7 = 36^7 - 26^7 = 78{,}364{,}164{,}096 - 8{,}031{,}810{,}176 = 70{,}332{,}353{,}920.$
  - $P_8 = 36^8 - 26^8 = 2{,}821{,}109{,}907{,}456 - 208{,}827{,}064{,}576$
    $= 2{,}612{,}282{,}842{,}880.$
- Consequently, $P = P_6 + P_7 + P_8 = 2{,}684{,}483{,}063{,}360.$

Internet Addresses

- Version 4 of the Internet Protocol (IPv4) uses 32 bits.



- **Class A Addresses** : used for the largest networks, a 0, followed by a 7-bit net_id and a 24-bit host_id.
- **Class B Addresses** : used for the medium-sized networks, a 10, followed by a 14-bit net_id and a 16-bit host_id.
- **Class C Addresses** : used for the smallest networks, a 110, followed by a 21-bit net_id and a 8-bit host_id.
  - Neither Class D nor Class E addresses are assigned as the address of a computer on the internet. Only Classes A, B, and C are available.
  - 1111111 is not available as the net_id of a Class A network.
  - Host_ids consisting of all 0s and all 1s are not available in any network.

Counting Internet Addresses

Example : How many different IPv4 addresses are available for computers on the internet?

Solution : Use both the sum and the product rule. Let x be the number of available addresses, and let x_A, x_B, and x_C denote the number of addresses for the respective classes.

- To find, x_A: $2^7 - 1 = 127$ net_ids. $2^{24} - 2 = 16{,}777{,}214$ host_ids.
  $x_A = 127 \cdot 16{,}777{,}214 = 2{,}130{,}706{,}178.$
- To find, x_B: $2^{14} = 16{,}384$ net_ids. $2^{16} - 2 = 16{,}534$ host_ids.
  $x_B = 16{,}384 \cdot 16{,}534 = 1{,}073{,}709{,}056.$
- To find, x_C: $2^{21} = 2{,}097{,}152$ net_ids. $2^8 - 2 = 254$ host_ids.
  $x_C = 2{,}097{,}152 \cdot 254 = 532{,}676{,}608.$
- Hence, the total number of available IPv4 addresses is

$$x = x_A + x_B + x_C$$
$$= 2{,}130{,}706{,}178 + 1{,}073{,}709{,}056 + 532{,}676{,}608$$
$$= 3{,}737{,}091{,}842.$$

Not Enough Today !!
The newer IPv6 protocol solves the problem of too few addresses.

Basic Counting Principles: Subtraction Rule

Subtraction Rule : If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

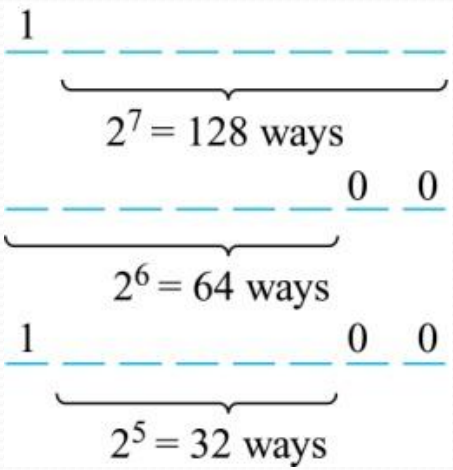
- Also known as, the *principle of inclusion-exclusion*:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Counting Bit Strings

Example : How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution : Use the subtraction rule.



- Number of bit strings of length eight that start with a 1 bit: $2^7 = 128$
- Number of bit strings of length eight that end with bits 00: $2^6 = 64$
- Number of bit strings of length eight that start with a 1 bit and end with bits 00 : $2^5 = 32$
  - Hence, the number is $128 + 64 - 32 = 160$.